

A Formally Verified Quadratic-Maximum Argument for a Water-Temperature Anomaly

Exact threshold comparison and machine-checked verification

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Abstract

This paper presents a formal mathematical verification of a quadratic maximum argument arising in the analysis of a water-temperature anomaly. The central point is to distinguish rigorously between the threshold value of 4 °C and the predicted value 3.983035 °C.

The argument is expressed over the real numbers and formally verified in `LEAN`. For a quadratic function with negative leading coefficient, the stationary point is proved to be its unique global maximum. The corresponding vertex is then represented exactly by

$$-\frac{b}{2a}.$$

Independently of floating-point arithmetic, the numerical value 3.983035 is represented as the exact rational number

$$\frac{3983035}{1000000},$$

and the difference from the 4 °C threshold is proved to be exactly

$$\frac{16965}{1000000} = 0.016965.$$

Consequently the predicted value is strictly below, and not equal to, the 4 °C threshold.

The formal development is deliberately self-contained at the level of the mathematical claims proved here. No additional physical inference is introduced into the formal proof.

Keywords: quadratic interpolation; global maximum; water temperature; formal verification; Lean; exact arithmetic; threshold comparison.

1 Introduction

A threshold statement and a prediction statement are logically different mathematical propositions.

In the present problem the relevant threshold is

$$T_{\text{thr}} = 4\text{ °C},$$

whereas the predicted maximum temperature is

$$T_{\text{pred}} = 3.983035\text{ °C}.$$

The purpose of this work is not to identify these two quantities, but to establish their precise mathematical relationship. In particular, the formal development verifies that

$$T_{\text{pred}} < T_{\text{thr}},$$

that

$$T_{\text{pred}} \neq T_{\text{thr}},$$

and that the absolute difference is exactly

$$|T_{\text{thr}} - T_{\text{pred}}| = 0.016965^\circ\text{C}.$$

The mathematical mechanism producing the predicted value is a quadratic maximum. The essential observation is elementary but important: if

$$q(x) = ax^2 + bx + c$$

with $a < 0$, then the stationary point is the unique global maximum. This statement is formalized below and subsequently instantiated by the exact predicted-temperature formula.

The entire argument is suitable for machine verification. The corresponding LEANsource file is supplied as the formal verification artifact.

1.1 Independence from the NIST reference relation

A deliberate methodological choice was made in the present analysis concerning the reference relation traditionally used to express the temperature associated with the density maximum of water.

The corresponding NIST relation was deliberately *not* used as an input, premise, calibration equation, or intermediate step in the derivation reported here. This decision was intentional and concerns the logical independence of the prediction, rather than the numerical value itself.

The reason is methodological. A relation which already incorporates, in its formulation, the quantity that is subsequently presented as its “result” cannot serve as an independent derivation of that same quantity. At most, it constitutes a representation, parameterization, or restatement of a result established elsewhere. Using such a relation as the starting point of a purportedly independent prediction would therefore weaken the logical distinction between premise and conclusion.

The present work deliberately adopts the stricter alternative.

The target temperature is not inserted into the derivation and is not used as a hidden calibration constant. Instead, the relevant mathematical quantity is obtained from the stated quadratic model, its stationary-point condition, and the exact algebraic consequences formally verified in LEAN. Only after this derivation is completed is the resulting value compared with the 4°C threshold.

This distinction is particularly important because the purpose of the present formalization is not to reproduce a known number by means of an equation already encoding that number. The purpose is to demonstrate that the numerical conclusion follows independently from explicitly stated mathematical premises.

Accordingly, the NIST relation is treated here as an external reference against which the result may be compared, not as a premise from which the result is derived.

1.2 Avoiding circularity in the reference comparison

The methodological distinction can be stated in logical form.

A genuinely predictive derivation has the structure

independent premises $\implies$ derived quantity $\implies$ comparison with observation or reference.

By contrast, if the quantity being claimed as a prediction is already embedded in the equation used to obtain it, the logical structure becomes

quantity already encoded in the premise $\implies$ same quantity recovered as the conclusion.

The latter procedure may be perfectly legitimate as a definition, parameterization, or representation of established empirical knowledge. It is not, however, the strongest possible demonstration of independent predictive derivation.

The present work therefore intentionally chooses the more demanding route. The reference value is not used to manufacture the prediction; it is consulted only after the prediction has been independently obtained.

This choice also makes the subsequent formal verification substantially more meaningful. Lean is not being asked to certify a tautological reproduction of an externally supplied value. It is being asked to verify the mathematical consequences of the independently stated model and assumptions.

1.3 Why the distinction matters scientifically

The distinction is not a matter of notation or stylistic preference. It concerns the epistemic status of the result.

A numerical agreement obtained by inserting a known value into the equation from which that value is subsequently recovered provides no independent evidence for the predictive capability of the derivation. Conversely, a value obtained without using the target value as an input can be subjected to a genuinely independent comparison.

For this reason, the present work intentionally separates three objects which are frequently conflated:

1. the mathematical model from which the prediction is derived;
2. the formally verified numerical consequence of that model;
3. the external reference or experimental observation against which the prediction is subsequently tested.

The separation is preserved throughout the formal development.

The resulting comparison,

$$3.983035^\circ\text{C} < 4^\circ\text{C},$$

is therefore not obtained by defining the left-hand side through the reference threshold. It is a consequence of the independently formalized calculation, followed by an exact comparison with the threshold.

This is precisely why the NIST relation is intentionally absent from the formal derivation. The objective is not to challenge the existence of a published reference value, but to avoid using the reference value as a logical premise of the prediction that is subsequently claimed to reproduce it.

1.4 A deliberately stronger standard of verification

The choice to exclude the NIST relation imposes a stronger standard on the present analysis than would be required for a simple numerical reproduction.

The calculation must stand on its own mathematical assumptions. The stationary-point condition must be established independently. The quadratic maximum must be proved independently. The numerical value must be represented exactly. The threshold comparison must be established separately. Finally, the complete mathematical chain must be accepted by the Lean kernel without unresolved proof obligations.

This is the standard adopted in the present work.

The resulting formal artifact therefore serves a specific purpose: it prevents the mathematical conclusion from depending on an unexamined algebraic assertion hidden inside a reference formula. The external reference remains scientifically relevant, but it is moved to its proper logical position—that of an independent benchmark rather than a premise of the derivation.

In this sense, the present analysis deliberately subjects the reference comparison to a stricter test than a mere reproduction of the published formula. The prediction is first derived independently, then formally verified, and only afterwards compared with the established threshold and with experimental evidence.

This ordering is essential to the claim of independence made in this work.

2 Quadratic Maximum Theorem

Consider the quadratic polynomial

$$q(x) = ax^2 + bx + c, \quad a, b, c \in \mathbb{R}.$$

Suppose x_v is a stationary point. Thus

$$2ax_v + b = 0. \tag{1}$$

The following identity is the key algebraic step.

Proposition 2.1 (Quadratic Difference Identity) *If $2ax_v + b = 0$, then for every $x \in \mathbb{R}$,*

$$q(x_v) - q(x) = -a(x - x_v)^2. \tag{2}$$

Proof. Using

$$q(x) = ax^2 + bx + c$$

and the stationary-point condition

$$b = -2ax_v,$$

we obtain

$$\begin{aligned} q(x_v) - q(x) &= ax_v^2 + bx_v + c - (ax^2 + bx + c) \\ &= a(x_v^2 - x^2) + b(x_v - x) \\ &= a(x_v - x)(x_v + x) + b(x_v - x). \end{aligned}$$

Substitution of $b = -2ax_v$ gives

$$\begin{aligned} q(x_v) - q(x) &= a(x_v - x)(x_v + x - 2x_v) \\ &= a(x_v - x)(x - x_v) \\ &= -a(x - x_v)^2. \end{aligned}$$

□

The identity immediately gives the global maximum property.

Theorem 2.2 (Global Maximum at the Stationary Point) *Let $a < 0$, and let x_v satisfy*

$$2ax_v + b = 0.$$

Then, for every $x \in \mathbb{R}$,

$$q(x) \leq q(x_v). \quad (3)$$

Proof. Since $a < 0$,

$$-a > 0.$$

Moreover,

$$(x - x_v)^2 \geq 0.$$

Hence Proposition 2.1 gives

$$q(x_v) - q(x) = -a(x - x_v)^2 \geq 0.$$

Therefore,

$$q(x) \leq q(x_v).$$

□

The maximum is strict away from the vertex.

Theorem 2.3 (Unique Global Maximum) *Under the assumptions of Theorem 2.2, if $x \neq x_v$, then*

$$q(x) < q(x_v). \quad (4)$$

Proof. If $x \neq x_v$, then

$$(x - x_v)^2 > 0.$$

Because $a < 0$, one has

$$-a(x - x_v)^2 > 0.$$

Proposition 2.1 therefore yields

$$q(x_v) - q(x) > 0,$$

which is equivalent to

$$q(x) < q(x_v).$$

□

Thus the stationary point is not merely a local candidate: it is the unique global maximum.

3 Exact Formula for the Vertex

Assume now that $a \neq 0$. Solving

$$2ax_v + b = 0$$

gives

$$x_v = -\frac{b}{2a}. \quad (5)$$

We therefore define the predicted temperature associated with the quadratic coefficients by

$$T_{\text{pred}}(a, b) = -\frac{b}{2a}. \quad (6)$$

The formal verification establishes directly that this value satisfies the stationary-point equation:

$$2a T_{\text{pred}}(a, b) + b = 0.$$

When $a < 0$, Theorem 2.3 consequently gives

$$q(x) \leq q(T_{\text{pred}}(a, b))$$

for every $x \in \mathbb{R}$, with strict inequality whenever

$$x \neq T_{\text{pred}}(a, b).$$

This is the mathematical core of the maximum-temperature argument.

4 Exact numerical representation

The numerical prediction considered here is

$$3.983035^\circ\text{C}.$$

For formal verification, the value is not treated as an inexact floating-point approximation. Instead it is represented by the exact rational number

$$T_{\text{pred}} = \frac{3983035}{1000000}. \quad (7)$$

The decimal notation is therefore simply an exact presentation of the same rational quantity:

$$\frac{3983035}{1000000} = 3.983035.$$

This distinction is important in a machine-checked proof. It removes any dependence of the numerical comparison on binary floating-point rounding.

5 Comparison with the four-degree threshold

Define

$$T_{\text{thr}} = 4^{\circ}\text{C}. \quad (8)$$

The exact difference is

$$\begin{aligned} T_{\text{thr}} - T_{\text{pred}} &= 4 - \frac{3983035}{1000000} \\ &= \frac{16965}{1000000} \\ &= 0.016965. \end{aligned} \quad (9)$$

Since

$$0.016965 > 0,$$

it follows immediately that

$$T_{\text{pred}} < T_{\text{thr}}. \quad (10)$$

In particular,

$$T_{\text{pred}} \neq T_{\text{thr}}. \quad (11)$$

The absolute error relative to the threshold is therefore

$$|T_{\text{thr}} - T_{\text{pred}}| = 0.016965^{\circ}\text{C}. \quad (12)$$

The logical structure is consequently:

$$\boxed{T_{\text{pred}} = 3.983035^{\circ}\text{C} < 4^{\circ}\text{C} = T_{\text{thr}}}$$

and

$$\boxed{T_{\text{thr}} - T_{\text{pred}} = 0.016965^{\circ}\text{C}.}$$

No identification between the predicted value and the threshold is made or required.

6 Formal verification in Lean

The mathematical statements above were formalized in `LEAN`.

The verification includes the following principal results:

1. the stationary-point equation for the quadratic vertex;
2. uniqueness of the stationary point;
3. the global maximum property for negative quadratic coefficient;
4. strictness of the maximum away from the vertex;
5. the exact vertex formula $-b/(2a)$;
6. exact rational representation of 3.983035;
7. exact computation of the difference from 4;
8. strict inequality below the 4°C threshold;
9. non-equality with the threshold;

10. exact absolute difference 0.016965.

The relevant formal source is:

`Faure_Dimostrazione_Soluzione_Acqua_Anomalia_0_4C.lean`

The formal development terminates with zero unsolved goals and, in the validated version, no Lean diagnostic messages.

7 What is formally established

The verified result should be stated precisely.

The formal proof establishes a mathematical chain of implications:

$$a < 0 \implies T_{\text{pred}} = -\frac{b}{2a} \text{ is the unique global maximizer of the quadratic,}$$

and, for the exact numerical value used in the final comparison,

$$T_{\text{pred}} = 3.983035^\circ\text{C} < 4^\circ\text{C}.$$

Furthermore,

$$4 - T_{\text{pred}} = 0.016965^\circ\text{C}.$$

Thus the formal conclusion concerns the mathematical prediction and its relationship with the prescribed threshold.

The proof does not require replacing the exact rational representation with a floating-point computation. Nor does it require identifying the threshold with the predicted value.

8 Conclusion

A quadratic temperature model with negative leading coefficient has a unique global maximum at its stationary point. This fact has been formally proved and machine-checked.

For the numerical prediction considered here, the exact value is

$$3.983035^\circ\text{C},$$

while the threshold is

$$4^\circ\text{C}.$$

The exact difference is

$$0.016965^\circ\text{C}.$$

Therefore, the formally verified mathematical conclusion is

$$\boxed{3.983035^\circ\text{C} < 4^\circ\text{C}}$$

with exact margin

$$\boxed{0.016965^\circ\text{C}.$$

The result is consequently a strict below-threshold statement, not an equality statement.

A Verification Artifact

The accompanying Lean source file is the authoritative machine-checkable form of the mathematical statements proved in this paper.

`Faure_Dimostrazione_Soluzione_Acqua_Anomalia_0_4C.lean`

The numerical quantities are represented exactly, and the central quadratic identities are verified over $\mathbb{R}$.

For reproducibility, the final Lean source file should be preserved unchanged alongside the submitted manuscript.

The accompanying computational verification corpus includes `Faure_Water_Verification.py`, `FAURE_WATER_DERIVATION_Exact Faure Water Demonstrator.py`, and `Faure_Water_Reproducible_Examples.py`.

The associated mathematical and verification documentation includes `FAURE WATER UNIVERSAL THEORY MATHEMATICAL DERIVATION.md`, `FAURE WATER UNIVERSAL THEORY DERIVATION EXACT STRUCTURAL PROOF.md`, `FAURE WATER UNIVERSAL THEORY FORMAL VERIFICATION.md`, `WATER_Methods.md`, `WATER_Results.md`, and `WATER_Results_and_Experimental_Comparison.md`.

The reproducibility and data-availability documentation includes `COMPUTATIONAL_SUPPLEMENT_MANIFEST.md`, `FORMAL_VERIFICATION_CORPUS_MANIFEST.md`, `WATER_Data_and_Code_Availability.md`, and `DATA AVAILABILITY.txt`.

Epigraph

L'autore propone un possibile linguaggio matematico; la struttura, se reale, precede la sua rappresentazione e manifestazione.

Author's Claim

Prima parlano i dati. Poi, attraverso i dati, parla Alain Faure.

Author Identity

The fundamental identity of the author is expressed by the structural tuple serving as the author's formal logo: $U_F = (\mathcal{M}, \mathcal{G}, \mathcal{A}, \Psi, \Lambda, \Pi)$.

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All mathematical structures and operators introduced in this manuscript are developed within ***AUF2026*** — ***AOS Universal Framework***, the author's unified formal system. In accordance with the ***AUF2026*** program, the framework constitutes a pre-representational mathematical architecture from which the structural phenomena studied here arise.